

Speech Synthesis BSAI – 7th Semester

Lab 9 Report

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Objective

To understand the process of speech synthesis by implementing a rule-based Grapheme-to-Phoneme (G2P) conversion system and introducing basic prosodic modeling through phoneme duration assignment.

Task 1.1: Grapheme-to-Phoneme (G2P) Conversion

Explanation: -to-Phoneme (G2P) conversion maps written text into a sequence of phonemes that can be pronounced by a Text-to-Speech (TTS) system. Since English spelling is irregular, rule-based mappings are applied to handle common pronunciation patterns. In this task, rules are defined for handling digraphs such as 'th', soft and hard 'c', double letters, and a small number of exceptions.

MATLAB Code:

```
1 function phoneme_sequence = simple_g2p(word)
2 % SIMPLE_G2P - A basic rule-based G2P converter for a small English
   subset.
3
4 if nargin < 1
5 error('You must provide a word');
6 end
7
8 word = lower(word);
9 phoneme_sequence = {};
10 i = 1;
11
12 while i <= length(word)
13 if i < length(word)
14 if strcmp(word(i:i+1), 'th')
15 phoneme_sequence = [phoneme_sequence, {'DH'}];
16 i = i + 2;
17 continue;
18 end
19 if word(i) == 'c'
20 if ismember(word(i+1), ['e','i','y'])
21 phoneme_sequence = [phoneme_sequence, {'S'}];
22 else
23 phoneme_sequence = [phoneme_sequence, {'K'}];
24 end
25 i = i + 1;
26 continue;
27 end
28 if (word(i) == word(i+1)) && isletter(word(i))
29 phoneme_sequence = [phoneme_sequence, {upper(word(i))}];
30 i = i + 2;
31 continue;
32 end
33 end
34 phoneme_sequence = [phoneme_sequence, {upper(word(i))}];
35 i = i + 1;
36 end
37
38 if strcmp(word, 'the')
39 phoneme_sequence = {'DH','AH'};
```

```

40 end
41 end
42
43 % Test words
44 words = {'cat','cent','apple','the','book','tree'};
45 for j = 1:length(words)
46 phonemes = simple_g2p(words{j});
47 disp(['Word: ', words{j}, ' -> Phonemes: ', strjoin(phonemes, '-')]);
48 end

```

```

>> words = {'cat','cent','apple','the','book','tree'};

for j = 1:length(words)
    phonemes = simple_g2p(words{j});
    disp(['Word: ', words{j}, ' -> Phonemes: ', strjoin(phonemes, '-')]);
end
Word: cat -> Phonemes: K-A-T
Word: cent -> Phonemes: S-E-N-T
Word: apple -> Phonemes: A-P-L-E
Word: the -> Phonemes: DH-AH
Word: book -> Phonemes: B-O-K
Word: tree -> Phonemes: T-R-E

```

Figure 1: Rule-based Grapheme-to-Phoneme Conversion Output

Task 1.2: Basic Prosodic Modeling – Duration

Explanation: refers to rhythm and timing characteristics of speech. To improve naturalness, phonemes are assigned durations based on their type. Vowels are given longer durations than consonants, and the final phoneme of a word is lengthened to simulate phrase-final effects.

MATLAB Code:

```

1 function [phoneme_sequence, durations] = simple_g2p_with_duration(word)
2 % SIMPLE_G2P_WITH_DURATION - Rule-based G2P + duration modeling.
3
4 word = lower(word);
5 phoneme_sequence = {};
6 durations = [];
7
8 vowels = 'aeiou';
9 base_vowel_duration = 100;      % ms
10 base_consonant_duration = 50;   % ms
11 final_syllable_extension = 20;  % ms
12
13 i = 1;
14 while i <= length(word)
15 phoneme = '';
16 if i < length(word)
17 if strcmp(word(i:i+1), 'th')
18 phoneme = 'DH';
19 i = i + 2;
20 elseif strcmp(word(i), 'c')
21 if any(strcmp(word(i+1), {'e','i','y'}))

```

```

22 phoneme = 'S';
23 else
24 phoneme = 'K';
25 end
26 i = i + 1;
27 elseif word(i) == word(i+1)
28 phoneme = upper(word(i));
29 i = i + 2;
30 else
31 phoneme = upper(word(i));
32 i = i + 1;
33 end
34 else
35 phoneme = upper(word(i));
36 i = i + 1;
37 end
38
39 '''
40 phoneme_sequence = [phoneme_sequence, {phoneme}];
41 if contains(lower(phoneme), vowels)
42     durations = [durations, base_vowel_duration];
43 else
44     durations = [durations, base_consonant_duration];
45 end
46 '''
47
48 end
49
50 if ~isempty(durations)
51 durations(end) = durations(end) + final_syllable_extension;
52 end
53
54 if strcmp(word, 'the')
55 phoneme_sequence = {'DH','AH'};
56 durations = [base_consonant_duration, base_vowel_duration +
57     final_syllable_extension];
58 end
59
60 % Test words
61 words = {'cat','apple','hello','the'};
62 for j = 1:length(words)
63 [p, d] = simple_g2p_with_duration(words{j});
64 disp(['Word: ', words{j}, ' -> Phonemes: ', strjoin(p,'-'), ...
65 ' -> Durations: ', num2str(d)]);
66 end

```

Lab Report Questions

1. Explain the process of G2P conversion and its importance in TTS. 2P conversion transforms written text into speech sounds. It is crucial because TTS systems cannot pronounce words directly from letters. Accurate phoneme mapping ensures intelligible and natural speech output.

2. **Challenges of rule-based G2P conversion:** -based systems struggle with irregular spellings, silent letters, and context-dependent pronunciations. The implemented

```

>> words = {'cat','apple','hello','the'};

for j = 1:length(words)
    [p, d] = simple_g2p_with_duration(words{j});
    disp(['Word: ', words{j}, ...
        ' -> Phonemes: ', strjoin(p, '-'), ...
        ' -> Durations: ', num2str(d)]);
end
Word: cat -> Phonemes: K-A-T -> Durations: 50 50 70
Word: apple -> Phonemes: A-P-L-E -> Durations: 50 50 50 70
Word: hello -> Phonemes: H-E-L-O -> Durations: 50 50 50 70
Word: the -> Phonemes: DH-AH -> Durations: 50 120
>>

```

Figure 2: Phoneme Duration Assignment Output

rules are limited in scope and do not cover stress, diphthongs, or complex phoneme combinations.

3. Improvement using advanced techniques: models and neural networks learn pronunciation patterns from large datasets, handle exceptions automatically, and achieve significantly higher accuracy than handcrafted rules.

4. Prosody in speech: includes pitch, duration, stress, and pauses. Modeling prosody is essential for natural-sounding speech, conveying meaning, emphasis, and emotion.

5. Description of duration model: are assigned longer durations than consonants, and the final phoneme is lengthened. Additional prosodic features include pitch contours, stress patterns, pauses, and energy variation.

Conclusion

This lab demonstrated the fundamental components of a Text-to-Speech system through rule-based G2P conversion and simple prosodic modeling. While limited in scope, the implementation highlights the importance of phonetic representation and timing in achieving intelligible and natural-sounding synthesized speech.